Klasifikasi Tingkat Keparahan Penyakit Alzheimer Berdasarkan Citra Otak

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https://github.com/rofiqsamanhudi/alzheimer-detection.git

Parameter

1. Machine Learning Klasik

- Pendekatan berbasis ekstraksi fitur hand-crafted + classifier tree-boosting.
- Fitur yang diekstrak:
- FFT (Fast Fourier Transform) — menggunakan scipy.fft.fft2
- GLCM (Gray-Level Co-occurrence Matrix) — via mahotas atau skimage
- HOG (Histogram of Oriented Gradients) — skimage.feature.hog
- Hu Moments — via cv2.HuMoments
- LBP (Local Binary Patterns) — uniform, skimage.feature.local_binary_pattern
- Wavelet Transform — pywt (db4, 3 level assumed)
- Zernike Moments — mahotas.features.zernike_moments (order 10 assumed)
- GIST descriptor — implementasi custom

Classifier:

- XGBoost (XGBClassifier, default parameters atau tuned via GridSearchCV; contoh: n_estimators=100, max_depth=6, learning_rate=0.1, objective='multi:softmax' dengan 4 kelas)

2. Convolutional Neural Networks (CNNs)

Parameter training umum CNN:

- Optimizer: Adam / AdamW (assumed)
- Loss: Categorical Crossentropy
- Batch size: 16–32 (assumed)
- Epochs: hingga 50 (dengan EarlyStopping)
- Learning rate: 0.0001–0.00001 (assumed, dengan scheduler seperti ReduceLROnPlateau)
- Augmentasi: rotasi ±10°, horizontal flip, zoom 0.1 (via torchvision.transforms atau ImageDataGenerator)

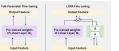
3. Transformer

Parameter training umum ViT:

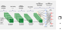
- Optimizer: AdamW
- Learning rate: 5e-5 → 1e-5 (fine-tune)
- Batch size efektif: 32 (dengan gradient accumulation)
- Epochs: hingga 10 (dengan EarlyStopping)
- Scheduler: CosineAnnealing + warmup

Deep Learning

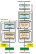
❒ Fine-Tuning



❒ CNN Pipeline



❒ Arsitektur Model Transformer



❒ PyTorch Framework

- Open-source deep learning library dari Facebook AI
- Graf komputasi dinamis untuk pembuatan model fleksibel
- GPU acceleration untuk pelatihan & inferensi efisien

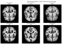
❒ Arsitektur Model Transformer

Model	Baseline	Fine-Tuning Normal	Fine-Tuning LoRA
CNN Scratch	95.29%	-	-
ResNet50	72.67%	78.47%	95.45%
EfficientNet	94.98%	93.09%	94.27%
Swin Transformer	93.71%	95.34%	94.73%
Efficient Former	95.14%	93.14%	94.73%
ViT B/16	96.26%	96.57%	92.78%

- **ViT B/16 + Fine-Tuning Normal** mencapai akurasi tertinggi 96.57%, karena mampu menangkap pola spasial global antar patch citra.
- **ResNet50 + LoRA** meraih akurasi 95.45%, dengan arsitektur residual yang stabil dan LoRA yang efisien mencegah overfitting.

Dataset

❒ Preprocessing Data



Tantangan

- Resize Stretch di 640×640
- Label Multi-Kelas
- Ukuran Dataset Besar (9.766 Gambar)
- Auto-Orientation & EXIF Stripping
- Sensitivitas Domain Medis

Rancangan Preprocessing

- Konversi warna (BGR → RGB)
- Image Resizing (224/224)
- LAB Color Space Transformation
- CLAHE Enhancement
- Normalisasi [0–1]
- Data Augmentation

❒ Dataset Benchmark

No Impairment	2.767 Gambar	28.33%
Very Mild Impairment	2.518 Gambar	25.78%
Mild Impairment	2.264 Gambar	23.18%
Moderate Impairment	2.217 Gambar	22.70%

Classic Feature Extraction

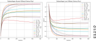
❒ Classic Feature Variant

Pendekatan yang digunakan:

- Fast Fourier Transform (FFT)
- Global Image Descriptor (GIST)
- Gray Level Co-occurrence Matrix (GLCM)
- Histogram of Oriented Gradients (HOG)
- Hu Moments
- Local Binary Pattern (LBP)
- Wavelet
- Zernike

❒ XG-Boost Pipeline

❒ Accuracy & loss



❒ Hasil Evaluasi

FFT	78.19%
GIST	92.93%
GLCM	86.38%
HOG	88.79%
Hui	88.58%
LBP	68.97%
Wavelet	76.70%
Zernike	80.34%

Catatan: *Fitur GIST memberikan performa terbaik di antara fitur klasik lainnya. GIST mampu menangkap informasi global dan tekstur citra dengan baik, sehingga XGBoost dapat memanfaatkan pola yang lebih representatif.*